

For example, if the measurement errors are distributed as a multivariate Gaussian, then equation (19.4.9) implies

$$\mathcal{A} = \frac{1}{2} \chi^2(\mathbf{c}, \mathbf{u}) \quad (19.4.15)$$

while a Bayesian prior expressing the belief that $\mathbf{u}$ should not have wild, large-amplitude oscillations might be captured by the multivariate Gaussian prior

$$\mathcal{B} = \lambda(\mathbf{u} \cdot \mathbf{u}) \quad (19.4.16)$$

Maximizing $P(\mathbf{u}|\mathbf{c}, I)$ to find the most probable $\hat{\mathbf{u}}$ is now exactly equivalent to equation (19.4.11). The constant λ is now merely a convenient parameterization for the narrowness of the Gaussian prior. It acts as a trade-off parameter, exactly as described above. In subsequent sections we will learn how to devise more sophisticated *smoothness priors*. In several cases these will be positive-definite quadratic forms in $\mathbf{u}$; those cases correspond to more complicated multivariate Gaussian priors on $\mathbf{u}$. In §19.7 we will meet a prior that is not exactly Gaussian, but rather based on the concept of *entropy*.

Within the Bayesian framework, you can do more than just solve for the most likely model $\hat{\mathbf{u}}$. For example, you can use Markov chain Monte Carlo (§15.8) to sample from the distribution of $\mathbf{u}$'s given the observed data.

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19.5 Linear Regularization Methods

What we will call *linear regularization* is also called the *Phillips-Twomey method* [1,2], the *constrained linear inversion method* [3], the *method of regularization* [4],

and *Tikhonov-Miller regularization* [5-7]. (It probably has other names also, since it is so obviously a good idea.) In its simplest form, the method is an immediate generalization of zeroth-order regularization (equation 19.4.11, above). As before, the functional $\mathcal{A}$ is taken to be the χ^2 deviation, equation (19.4.9), but the functional $\mathcal{B}$ is replaced by more sophisticated measures of smoothness that derive from first or higher derivatives.

For example, suppose that your a priori belief is that a credible $u(x)$ is not too different from a constant. Then a reasonable functional to minimize is

$$\mathcal{B} \propto \int [\hat{u}'(x)]^2 dx \propto \sum_{\mu=0}^{M-2} [\hat{u}_\mu - \hat{u}_{\mu+1}]^2 \quad (19.5.1)$$

since it is nonnegative and equal to zero only when $\hat{u}(x)$ is constant. Here $\hat{u}_\mu \equiv \hat{u}(x_\mu)$, and the second equality (proportionality) assumes that the x_μ 's are uniformly spaced. We can write the second form of $\mathcal{B}$ as

$$\mathcal{B} = |\mathbf{B} \cdot \hat{\mathbf{u}}|^2 = \hat{\mathbf{u}} \cdot (\mathbf{B}^T \cdot \mathbf{B}) \cdot \hat{\mathbf{u}} \equiv \hat{\mathbf{u}} \cdot \mathbf{H} \cdot \hat{\mathbf{u}} \quad (19.5.2)$$

where $\hat{\mathbf{u}}$ is the vector of components $\hat{u}_\mu$, $\mu = 0, \dots, M-1$; $\mathbf{B}$ is the $(M-1) \times M$ first difference matrix

$$\mathbf{B} = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \ddots & & & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix} \quad (19.5.3)$$

and $\mathbf{H}$ is the $M \times M$ matrix

$$\mathbf{H} = \mathbf{B}^T \cdot \mathbf{B} = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \ddots & & & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix} \quad (19.5.4)$$

Note that $\mathbf{B}$ has one fewer row than column. It follows that the symmetric $\mathbf{H}$ is degenerate; it has exactly one zero eigenvalue corresponding to the *value* of a constant function, any one of which makes $\mathcal{B}$ exactly zero.

If, just as in §15.4, we write

$$A_{i\mu} \equiv R_{i\mu}/\sigma_i \quad b_i \equiv c_i/\sigma_i \quad (19.5.5)$$

then, using equation (19.4.9), the minimization principle (19.4.12) is

$$\text{minimize: } \mathcal{A} + \lambda \mathcal{B} = |\mathbf{A} \cdot \hat{\mathbf{u}} - \mathbf{b}|^2 + \lambda \hat{\mathbf{u}} \cdot \mathbf{H} \cdot \hat{\mathbf{u}} \quad (19.5.6)$$

This can readily be reduced to a linear set of *normal equations*, just as in §15.4: The components $\hat{u}_\mu$ of the solution satisfy the set of M equations in M unknowns,

$$\sum_{\rho} \left[\left(\sum_i A_{i\mu} A_{i\rho} \right) + \lambda H_{\mu\rho} \right] \hat{u}_\rho = \sum_i A_{i\mu} b_i \quad \mu = 0, 1, \dots, M-1 \quad (19.5.7)$$

or, in vector notation,

$$(\mathbf{A}^T \cdot \mathbf{A} + \lambda \mathbf{H}) \cdot \hat{\mathbf{u}} = \mathbf{A}^T \cdot \mathbf{b} \quad (19.5.8)$$

Equations (19.5.7) or (19.5.8) can be solved by the standard techniques of Chapter 2, e.g., LU decomposition. The usual warnings about normal equations being ill-conditioned do not apply, since the whole purpose of the λ term is to cure that same ill-conditioning. Note, however, that the λ term *by itself* is ill-conditioned, since it does not select a preferred constant value. You hope your data can at least do *that*!

Although inversion of the matrix $(\mathbf{A}^T \cdot \mathbf{A} + \lambda \mathbf{H})$ is not generally the best way to solve for $\hat{\mathbf{u}}$, let us digress to write the solution to equation (19.5.8) schematically as

$$\hat{\mathbf{u}} = \left(\frac{1}{\mathbf{A}^T \cdot \mathbf{A} + \lambda \mathbf{H}} \cdot \mathbf{A}^T \cdot \mathbf{A} \right) \mathbf{A}^{-1} \cdot \mathbf{b} \quad (\text{schematic only!}) \quad (19.5.9)$$

where the identity matrix in the form $\mathbf{A} \cdot \mathbf{A}^{-1}$ has been inserted. This is schematic not only because the matrix inverse is fancifully written as a denominator, but also because, in general, the inverse matrix $\mathbf{A}^{-1}$ does not exist. However, it is illuminating to compare equation (19.5.9) with equation (13.3.6) for optimal or Wiener filtering, or with equation (13.6.6) for general linear prediction. (The concepts of §15.9 are also related.) One sees that $\mathbf{A}^T \cdot \mathbf{A}$ plays the role of S^2 , the signal power or autocorrelation, while $\lambda \mathbf{H}$ plays the role of N^2 , the noise power or autocorrelation. The term in parentheses in equation (19.5.9) is something like an optimal filter, whose effect is to pass the ill-posed inverse $\mathbf{A}^{-1} \cdot \mathbf{b}$ through unmodified when $\mathbf{A}^T \cdot \mathbf{A}$ is sufficiently large, but to suppress it when $\mathbf{A}^T \cdot \mathbf{A}$ is small.

The above choices of $\mathbf{B}$ and $\mathbf{H}$ are only the simplest in an obvious sequence of derivatives. If your a priori belief is that a *linear* function is a good approximation to $u(x)$, then minimize

$$\mathcal{B} \propto \int [\hat{u}''(x)]^2 dx \propto \sum_{\mu=0}^{M-3} [-\hat{u}_{\mu} + 2\hat{u}_{\mu+1} - \hat{u}_{\mu+2}]^2 \quad (19.5.10)$$

implying

$$\mathbf{B} = \begin{pmatrix} -1 & 2 & -1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \ddots & & & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & -1 & 2 & -1 \end{pmatrix} \quad (19.5.11)$$

and

$$\mathbf{H} = \mathbf{B}^T \cdot \mathbf{B} = \begin{pmatrix} 1 & -2 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ -2 & 5 & -4 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 1 & -4 & 6 & -4 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & -4 & 6 & -4 & 1 & 0 & \cdots & 0 \\ \vdots & & & & \ddots & & & & \vdots \\ 0 & \cdots & 0 & 1 & -4 & 6 & -4 & 1 & 0 \\ 0 & \cdots & 0 & 0 & 1 & -4 & 6 & -4 & 1 \\ 0 & \cdots & 0 & 0 & 0 & 1 & -4 & 5 & -2 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 1 & -2 & 1 \end{pmatrix} \quad (19.5.12)$$

This $\mathbf{H}$ has two zero eigenvalues, corresponding to the two undetermined parameters of a linear function.

If your a priori belief is that a *quadratic* function is preferable, then minimize

$$\mathcal{B} \propto \int [\hat{u}'''(x)]^2 dx \propto \sum_{\mu=0}^{M-4} [-\hat{u}_{\mu} + 3\hat{u}_{\mu+1} - 3\hat{u}_{\mu+2} + \hat{u}_{\mu+3}]^2 \quad (19.5.13)$$

with

$$\mathbf{B} = \begin{pmatrix} -1 & 3 & -3 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 3 & -3 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \ddots & & & & \vdots \\ 0 & \cdots & 0 & 0 & -1 & 3 & -3 & 1 & 0 \\ 0 & \cdots & 0 & 0 & 0 & -1 & 3 & -3 & 1 \end{pmatrix} \quad (19.5.14)$$

and now

$$\mathbf{H} = \begin{pmatrix} 1 & -3 & 3 & -1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ -3 & 10 & -12 & 6 & -1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 3 & -12 & 19 & -15 & 6 & -1 & 0 & 0 & 0 & \cdots & 0 \\ -1 & 6 & -15 & 20 & -15 & 6 & -1 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 6 & -15 & 20 & -15 & 6 & -1 & 0 & \cdots & 0 \\ \vdots & & & & \ddots & & & & \vdots & & \vdots \\ 0 & \cdots & 0 & -1 & 6 & -15 & 20 & -15 & 6 & -1 & 0 \\ 0 & \cdots & 0 & 0 & -1 & 6 & -15 & 20 & -15 & 6 & -1 \\ 0 & \cdots & 0 & 0 & 0 & -1 & 6 & -15 & 19 & -12 & 3 \\ 0 & \cdots & 0 & 0 & 0 & 0 & -1 & 6 & -12 & 10 & -3 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & -1 & 3 & -3 & 1 \end{pmatrix} \quad (19.5.15)$$

(We'll leave the calculation of cubics and above to the compulsive reader.)

Notice that you can regularize with “closeness to a differential equation,” if you want. Just pick $\mathbf{B}$ to be the appropriate sum of finite difference operators (the coefficients can depend on x) and calculate $\mathbf{H} = \mathbf{B}^T \cdot \mathbf{B}$. You don't need to know the values of your boundary conditions, since $\mathbf{B}$ can have fewer rows than columns, as above; hopefully, your data will determine them. Of course, if you do know some boundary conditions, you can build these into $\mathbf{B}$, too.

With all the proportionality signs above, you may have lost track of what actual value of λ to try first. A simple trick for at least getting “on the map” is to first try

$$\lambda = \text{Tr}(\mathbf{A}^T \cdot \mathbf{A}) / \text{Tr}(\mathbf{H}) \quad (19.5.16)$$

where Tr is the trace of the matrix (sum of diagonal components). This choice will tend to make the two parts of the minimization have comparable weights, and you can adjust from there.

As for what is the “correct” value of λ , an objective criterion, if you know your errors σ_i with reasonable accuracy, is to make χ^2 (that is, $|\mathbf{A} \cdot \hat{\mathbf{u}} - \mathbf{b}|^2$) equal to N , the number of measurements. We remarked above on the twin acceptable

choices $N \pm (2N)^{1/2}$. A subjective criterion is to pick any value that you like in the range $0 < \lambda < \infty$, depending on your relative degree of belief in the a priori and a posteriori evidence. (Yes, people actually do that.) The problem with being a rigorous Bayesian at this stage is that rarely, if ever, is your understanding of the prior so complete as to give a firm, objective, value for λ , and, as we pointed out in §15.1.1, purely Bayesian methods for assessing goodness-of-fit are largely nonexistent.

19.5.1 Two-Dimensional Problems and Iterative Methods

Up to now our notation has been indicative of a one-dimensional problem, finding $\hat{u}(x)$ or $\hat{u}_\mu = \hat{u}(x_\mu)$. However, all of the discussion easily generalizes to the problem of estimating a two-dimensional set of unknowns $\hat{u}_{\mu\kappa}$, $\mu = 0, \dots, M-1$, $\kappa = 0, \dots, K-1$, corresponding, say, to the pixel intensities of a measured image. In this case, equation (19.5.8) is still the one we want to solve.

In image processing, it is usual to have the same number of input pixels in a measured “raw” or “dirty” image as desired “clean” pixels in the processed output image, so the matrices $\mathbf{R}$ and $\mathbf{A}$ (equation 19.5.5) are square and of size $MK \times MK$. $\mathbf{A}$ is typically too large to represent as a full matrix, but often it is either (i) sparse, with coefficients blurring an underlying pixel (i, j) only into measurements $(i \pm \text{few}, j \pm \text{few})$, or (ii) translationally invariant, so that $A_{(i,j)(\mu,\nu)} = A(i - \mu, j - \nu)$. Both of these situations lead to tractable problems.

In the case of translational invariance, fast Fourier transforms (FFTs) are the obvious method of choice. The general linear relation between underlying function and measured values (19.4.7) now becomes a discrete convolution like equation (13.1.1). If $\mathbf{k}$ denotes a two-dimensional wave vector, then the two-dimensional FFT takes us back and forth between the transform pairs

$$A(i - \mu, j - \nu) \iff \tilde{\mathbf{A}}(\mathbf{k}) \quad b_{(i,j)} \iff \tilde{b}(\mathbf{k}) \quad \hat{u}_{(i,j)} \iff \tilde{u}(\mathbf{k}) \quad (19.5.17)$$

We also need a regularization or smoothing operator $\mathbf{B}$ and the derived $\mathbf{H} = \mathbf{B}^T \cdot \mathbf{B}$. One popular choice for $\mathbf{B}$ is the five-point finite difference approximation of the Laplacian operator, that is, the difference between the value of each point and the average of its four Cartesian neighbors. In Fourier space, this choice implies

$$\begin{aligned} \tilde{B}(\mathbf{k}) &\propto \sin^2(\pi k_1/M) \sin^2(\pi k_2/K) \\ \tilde{H}(\mathbf{k}) &\propto \sin^4(\pi k_1/M) \sin^4(\pi k_2/K) \end{aligned} \quad (19.5.18)$$

In Fourier space, equation (19.5.7) is merely algebraic, with solution

$$\tilde{u}(\mathbf{k}) = \frac{\tilde{A}^*(\mathbf{k})\tilde{b}(\mathbf{k})}{|\tilde{A}(\mathbf{k})|^2 + \lambda \tilde{H}(\mathbf{k})} \quad (19.5.19)$$

where the asterisk denotes complex conjugation. You can make use of the FFT routines for real data in §12.6.

Turn now to the case where $\mathbf{A}$ is not translationally invariant. Direct solution of (19.5.8) is now hopeless, since the matrix $\mathbf{A}$ is just too large. We need some kind of iterative scheme.

One way to proceed is to use the full machinery of the conjugate gradient method in §10.8 to find the minimum of $\mathcal{A} + \lambda\mathcal{B}$, equation (19.5.6). Of the various methods in Chapter 10, the conjugate gradient is the unique best choice because (i) it does not require storage of a Hessian matrix, which would be infeasible here, and (ii) it does exploit gradient information, which we can readily compute: The gradient of equation (19.5.6) is

$$\nabla(\mathcal{A} + \lambda\mathcal{B}) = 2[(\mathbf{A}^T \cdot \mathbf{A} + \lambda\mathbf{H}) \cdot \hat{\mathbf{u}} - \mathbf{A}^T \cdot \mathbf{b}] \quad (19.5.20)$$

(cf. 19.5.8). Evaluation of both the function and the gradient should of course take advantage of the sparsity of $\mathbf{A}$, for example using the methods `ax` and `atx` in the `NRsparseMat` object in §2.7.5. We will discuss the conjugate gradient technique further in §19.7, in the context of the (nonlinear) maximum entropy method. Some of that discussion can apply here as well.

The conjugate gradient method notwithstanding, application of the unsophisticated steepest descent method (see §10.8) can sometimes produce useful results, particularly when combined with projections onto convex sets (see below). If the solution after k iterations is denoted $\hat{\mathbf{u}}^{(k)}$, then after $k + 1$ iterations we have

$$\hat{\mathbf{u}}^{(k+1)} = [\mathbf{1} - \epsilon(\mathbf{A}^T \cdot \mathbf{A} + \lambda\mathbf{H})] \cdot \hat{\mathbf{u}}^{(k)} + \epsilon\mathbf{A}^T \cdot \mathbf{b} \quad (19.5.21)$$

Here ϵ is a parameter that dictates how far to move in the downhill gradient direction. The method converges when ϵ is small enough, in particular satisfying

$$0 < \epsilon < \frac{2}{\max \text{eigenvalue}(\mathbf{A}^T \cdot \mathbf{A} + \lambda\mathbf{H})} \quad (19.5.22)$$

There exist complicated schemes for finding optimal values or sequences for ϵ , see [7]; or, one can adopt an experimental approach, evaluating (19.5.6) to be sure that downhill steps are in fact being taken.

In those image processing problems where the final measure of success is somewhat subjective (e.g., “how good does the picture look?”), iteration (19.5.21) sometimes produces significantly improved images long before convergence is achieved. This probably accounts for much of its use, since its mathematical convergence is extremely slow. In fact, (19.5.21) can be used with $\mathbf{H} = 0$, in which case the solution is not regularized at all, and full convergence would be disastrous! This is called *Van Cittert’s method* and goes back to the 1930s. A number of iterations the order of 10^3 is not uncommon [7].

19.5.2 Deterministic Constraints: Projections onto Convex Sets

A set of possible underlying functions (or images) $\{\hat{\mathbf{u}}\}$ is said to be *convex* if, for any two elements $\hat{\mathbf{u}}_a$ and $\hat{\mathbf{u}}_b$ in the set, all the linearly interpolated combinations

$$(1 - \eta)\hat{\mathbf{u}}_a + \eta\hat{\mathbf{u}}_b \quad 0 \leq \eta \leq 1 \quad (19.5.23)$$

are also in the set. Many *deterministic constraints* that one might want to impose on the solution $\hat{\mathbf{u}}$ to an inverse problem in fact define convex sets, for example:

- positivity
- compact support (i.e., zero value outside of a certain region)
- known bounds (i.e., $u_L(x) \leq \hat{u}(x) \leq u_U(x)$ for specified functions u_L and u_U).

(In this last case, the bounds might be related to an initial estimate and its error bars, e.g., $\hat{u}_0(x) \pm \gamma\sigma(x)$, where γ is of order 1 or 2.) Notice that these, and similar, constraints can be either in the image space, or in the Fourier transform space, or (in fact) in the space of any linear transformation of $\hat{\mathbf{u}}$.

If C_i is a convex set, then $\mathcal{P}_i$ is called a *nonexpansive projection operator* onto that set if (i) $\mathcal{P}_i$ leaves unchanged any $\hat{\mathbf{u}}$ already in C_i , and (ii) $\mathcal{P}_i$ maps any $\hat{\mathbf{u}}$ outside C_i to the *closest* element of C_i , in the sense that

$$|\mathcal{P}_i \hat{\mathbf{u}} - \hat{\mathbf{u}}| \leq |\hat{\mathbf{u}}_a - \hat{\mathbf{u}}| \quad \text{for all } \hat{\mathbf{u}}_a \text{ in } C_i \quad (19.5.24)$$

While this definition sounds complicated, examples are very simple: A nonexpansive projection onto the set of positive $\hat{\mathbf{u}}$'s is "set all negative components of $\hat{\mathbf{u}}$ equal to zero." A nonexpansive projection onto the set of $\hat{u}(x)$'s bounded by $u_L(x) \leq \hat{u}(x) \leq u_U(x)$ is "set all values less than the lower bound equal to that bound, and set all values greater than the upper bound equal to *that* bound." A nonexpansive projection onto functions with compact support is "zero the values outside of the region of support."

The usefulness of these definitions is the following remarkable theorem: Let C be the intersection of m convex sets $C_1, C_2, \dots, C_m$. Then the iteration

$$\hat{\mathbf{u}}^{(k+1)} = (\mathcal{P}_1 \mathcal{P}_2 \dots \mathcal{P}_m) \hat{\mathbf{u}}^{(k)} \quad (19.5.25)$$

will converge to C from all starting points, as $k \rightarrow \infty$. Also, if C is empty (there is no intersection), then the iteration will have no limit point. Application of this theorem is called the *method of projections onto convex sets* or sometimes *POCS* [7].

A generalization of the POCS theorem is that the $\mathcal{P}_i$'s can be replaced by a set of $\mathcal{T}_i$'s,

$$\mathcal{T}_i \equiv \mathbf{1} + \beta_i(\mathcal{P}_i - \mathbf{1}) \quad 0 < \beta_i < 2 \quad (19.5.26)$$

A well-chosen set of β_i 's can accelerate the convergence to the intersection set C .

Some inverse problems can be completely solved by iteration (19.5.25) alone! For example, a problem that occurs in both astronomical imaging and X-ray diffraction work is to recover an image given only the *modulus* of its Fourier transform (equivalent to its power spectrum or autocorrelation) and not the *phase*. Here two convex sets can be utilized: the set of all positive images, and the set of all images with zero intensity outside of some specified region. A third set, the set of all images whose Fourier transform has the specified modulus to within specified error bounds, is not convex: It is an annulus in the complex plane for each Fourier component. (Of course FFTs are used to get in and out of Fourier space each time the Fourier constraint is imposed.) The POCS iteration (19.5.25) that cycles among all three sets, imposing each constraint in turn, is thus not guaranteed (by the POCS theorem) to converge; it can get stuck in *traps* [8]. However, it often does work.

The specific application of POCS to constraints (not necessarily all convex) alternately in the spatial and Fourier domains is known as the *Gerchberg-Saxton* algorithm [9]. While this algorithm is nonexpansive, and is frequently convergent

in practice, it does not converge in all cases [8,10]. In the phase-retrieval problem just mentioned, the algorithm often gets stuck in traps for many iterations. After as many as 10^4 to 10^5 iterations, sudden, dramatic improvements may occur. In principle, some traps may be permanent, requiring more interventional “unsticking” procedures; see [8,11]. The uniqueness of the solution is also not well understood, although for two-dimensional images of reasonable complexity it is believed to be unique. The use of nonconvex sets in an iteration like (19.5.25) is called the *method of generalized projections*.

Deterministic constraints can be incorporated, via projection operators, into iterative methods of linear regularization. In particular, rearranging terms somewhat, we can write the iteration (19.5.21) as

$$\hat{\mathbf{u}}^{(k+1)} = (\mathbf{I} - \epsilon\lambda\mathbf{H}) \cdot \hat{\mathbf{u}}^{(k)} + \epsilon\mathbf{A}^T \cdot (\mathbf{b} - \mathbf{A} \cdot \hat{\mathbf{u}}^{(k)}) \quad (19.5.27)$$

If the iteration is modified by the insertion of projection operators at each step

$$\hat{\mathbf{u}}^{(k+1)} = (\mathcal{P}_1\mathcal{P}_2 \cdots \mathcal{P}_m)[(\mathbf{I} - \epsilon\lambda\mathbf{H}) \cdot \hat{\mathbf{u}}^{(k)} + \epsilon\mathbf{A}^T \cdot (\mathbf{b} - \mathbf{A} \cdot \hat{\mathbf{u}}^{(k)})] \quad (19.5.28)$$

(or, instead of $\mathcal{P}_i$'s, the $\mathcal{T}_i$ operators of equation 19.5.26), then it can be shown that the convergence condition (19.5.22) is unmodified, and the iteration will converge to minimize the quadratic functional (19.5.6) subject to the desired nonlinear deterministic constraints. See [7] for references to more sophisticated, and faster converging, iterations along these lines.

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